

Project Proposal (Proposed Solution) Report

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

The proposal aims to bring Machine Learning to flood risk assessment, replacing slow, manual forecasting with a fast, data-driven web application that gives residents and authorities an instant flood-risk prediction.

Project Overview

Field	Details
Objective	To predict flood occurrence from historical rainfall and weather data using a Machine Learning model, and deliver the prediction through an easy-to-use web application.
Scope	Covers data collection and preprocessing, training a Random Forest classifier, and deploying it behind a Flask web app with a simple HTML form for real-time predictions.

Problem Statement

Field	Details
Description	Traditional flood forecasting relies on manual meteorological analysis, which is slow and lacks location-specific, on-demand predictions for residents.
Impact	Delayed or generic warnings reduce preparation time and increase risk to life and property during flood events.

Proposed Solution

Field	Details
Approach	Train a Random Forest Classifier on historical rainfall and weather data, then serve it through a Flask web application for instant, on-demand flood-risk predictions.
Key Features	- ML-based flood risk prediction from weather inputs. - Simple web form for entering weather parameters. - Instant, real-time prediction display. - Reusable, retrainable model pipeline (train_model.py).

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU specifications	Standard multi-core CPU (no GPU required)

Resource Type	Description	Specification / Allocation
Memory	RAM specifications	4–8 GB
Storage	Disk space for data, model, and logs	< 500 MB
Frameworks	Python web framework	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, joblib, openpyxl
Development Environment	IDE	Jupyter Notebook, Visual Studio Code
Data	Source, size, format	Kaggle Rainfall Dataset, .xlsx format